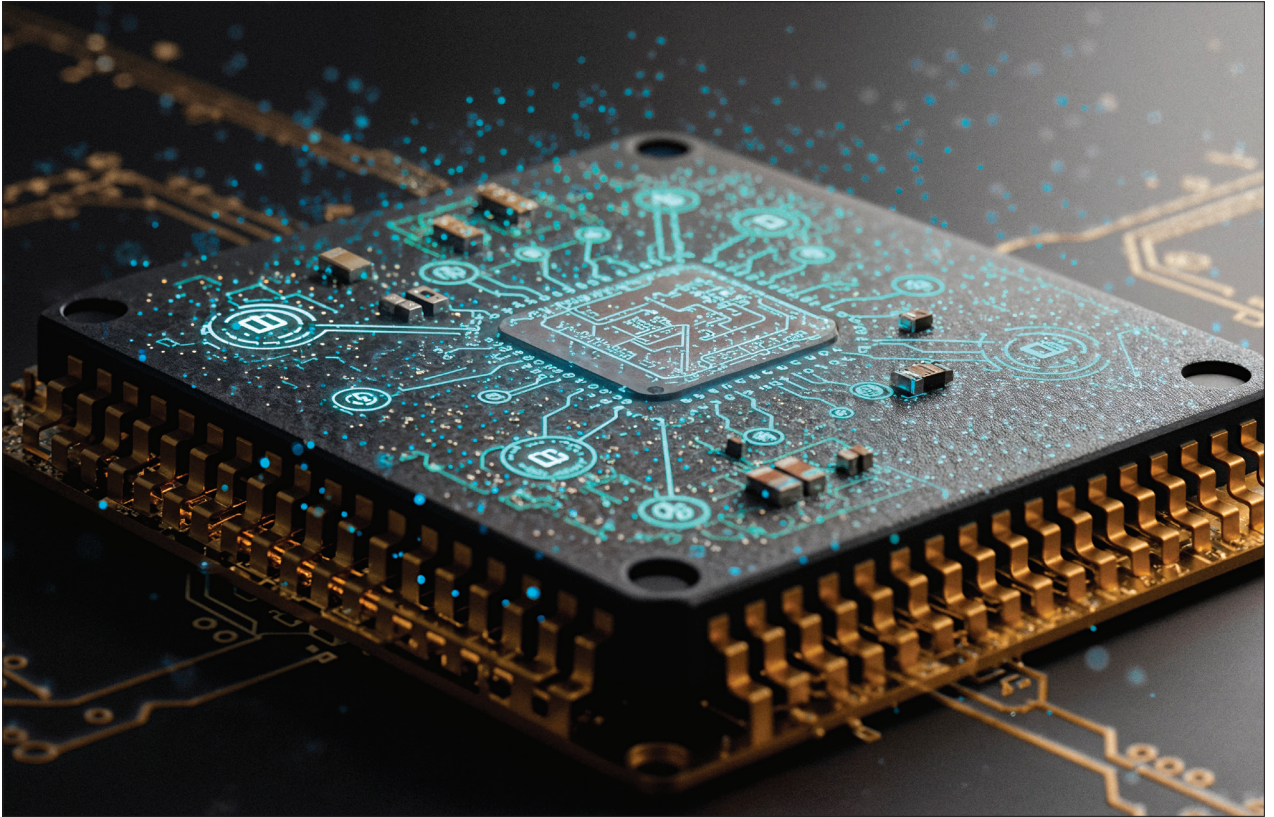




Chips are tiny cities

From sand to silicon, how chips come alive

Let's begin by killing the most common myth in consumer tech: a "chip" is not *one thing*. It's not "the CPU." It's not "the processor." It's not even necessarily one piece of silicon.

A modern chip is closer to a tiny city. It has neighborhoods (functional blocks), roads (interconnects), power stations (power delivery), traffic rules (timing), and a mayor (the clock) that decides when things are allowed to happen. And, like any city, its performance is dictated not by one superstar building, but by how everything connects, moves, and stays within limits – power, heat, cost, and reliability.

This chapter gives you the mental model you'll use for the rest of the book: *what chips are made of, why "nm" numbers can be misleading, why heat is the real villain, and why packaging is now a performance feature.*

What exactly is a chip

A chip (integrated circuit) is a huge number of electronic components – mainly transistors – built on a single piece of semiconductor material to perform specific functions.

That's it. That's the whole story. Everything else is just scale, cleverness, and trade-offs.

The reason chips feel like magic is the number of transistors involved. Moore's Law – often summarized as transistor counts doubling roughly every two years – captures how chip complexity kept exploding for decades. But Moore's Law isn't a law of physics; it's an industrial drumbeat powered by manufacturing learning curves, better tools, better design, and insane capital investment.

Key idea: a chip is not "faster because it's smaller." A chip is faster (or more efficient) because engineers squeeze in *more*